# Sheikh Mohsin

B.Sc. Computer Science & Business Analytics, FLAME University · Pune, India · Research internships

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## Education

**FLAME University, Pune** · *B.Sc. CS & Business Analytics, full scholarship · CGPA 7.65/10* Aug 2024 – Present

- Modeled and verified concurrent and distributed systems with SAT solvers, temporal-logic model checking, and Alloy/TLA+ specifications
- Implemented and compared classification and clustering methods, using PCA to test how feature selection affected model behavior
- Term paper on globalization and India's LPG-era economic reforms (Public Policy coursework)

**Jawahar Navodaya Vidyalaya, Ganderbal** · *Senior Secondary Education, merit scholarship, residential* 2017 – 2024

- Coordinated Atal Tinkering Labs and taught junior students to turn ideas into 3D-printed prototypes

## Research Experience

**IIT Ropar** · *Research Intern (Summer Internship Program), SSD Systems* May 2026 – Present

- Learned SSD storage systems from first principles under Prof. Venkata Kalyan Tavva and compared recency- and frequency-based cache eviction strategies in the SimpleSSD-Standalone 2.0 simulator as a foundation for an original performance study
- Designed and implemented spatial prefetching that predicts nearby address-translation entries from how data is grouped in flash; found and fixed a cache-order bug that caused newly loaded entries to evict one another before use
- Investigated how ransomware write bursts interact with the drive's internal cleanup, identified a race that may destroy recoverable data before detection responds, and assessed whether a targeted defense could form a new research contribution

**Discover India Program, FLAME University** · *Research, Taraksi (Silver Filigree)* Sept 2025 – Apr 2026

- Studied, within a 13-member team, how Taraksi silver filigree moved from a household craft to a commercial practice
- Traced the craft's history and technique through literature, then designed field surveys and semi-structured interviews with artisans
- Synthesized qualitative findings into academic documentation and cultural analysis

**Goonj, Delhi** · *Research and Operations Intern* May 2025 – Jul 2025

- Traced donation and redistribution data through the sorting pipeline and tested data patterns against field-team observations before proposing process changes

## Projects

**Cache Performance Benchmark Suite** · *C++, `perf_event_open`, RDTSCP* Repository

- Measured hardware clock-cycle costs across 16 data structures and compared contiguous and pointer-based layouts across access patterns and data sizes

**IPL Data Mining** · *Python* Repository

- Repeatable analysis workflow that separates data preparation from interpretation so conclusions can be checked against the records

## Writing

- Co-author, "Lost in Instruction", Deccan Herald, on education policy and technology
- Reflective and literary writing, including a published poem, at Kashmir Poetics; essays at Feelings and Numbers (Substack)

## Skills

**Programming:** Python, C++, JavaScript, SQL, LaTeX, Git/GitHub, Linux
**Systems and data:** FTL/NAND flash internals, hardware performance profiling (`perf_event_open`, RDTSCP), data analysis, data visualization, statistical reasoning
**Communication:** Technical writing, research communication, survey and interview design; English, Hindi, Urdu, Kashmiri

## Leadership

**Karta Initiative India Foundation** · *University Leader (40+ scholars); Technical and Operations Lead, workshops* Aug 2024 – Present